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# Nautilus Layout Engine

A Platypus-inspired flow layout for Go PDF generation.

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## 1. Introduction

The layout engine builds on top of the Nautilus pdf.Document drawing primitives and adds a higher-level abstraction: the Flowable interface. Any piece of content — a paragraph, an image, a spacer, a horizontal rule — implements Flowable and can be placed in a Story.

The DocTemplate engine processes the story sequentially. When a frame fills up, it automatically advances to the next frame or starts a new page, calling user-supplied decorators to render headers and footers along the way.

This mirrors the design of Platypus from the Python ReportLab library, adapted to Go idioms: explicit struct initialization instead of keyword arguments, interfaces instead of class inheritance, and value semantics where appropriate.

## 2. Flowable Types

The following built-in Flowable types are provided:

### Paragraph

The primary text element. Supports word-wrapping, font and colour overrides, left/right indentation, and horizontal alignment (left, centre, right). Long paragraphs can be split across frames.

### Spacer

Reserves a fixed amount of vertical space without rendering anything visible. Useful for adding breathing room between sections.

### HRFlowable

Draws a horizontal rule as a solid filled bar. Width can be an absolute point value or a fraction of the available width (e.g. 0.8 = 80 %). Horizontal alignment is configurable.

### KeepTogether

Wraps a group of flowables that must appear on the same frame. If the group does not fit in the remaining space, the engine inserts a FrameBreak and retries on the next frame. If the group is larger than an entire frame, it falls back to individual splitting.

## 3. Action Flowables

Action flowables are zero-height elements that control the engine rather than rendering visible content.

PageBreak forces an immediate page break. An optional NextTemplate ID switches the page template on the new page. FrameBreak advances to the next frame within the current page. CondPageBreak inserts a page break only when fewer than MinHeight points remain in the current frame. NextPageTemplate schedules a template switch that takes effect on the next page break.

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## 4. Page Templates

A PageTemplate groups one or more LayoutFrames with optional OnPage / OnPageEnd decorators. The engine fills frames in the order they are listed; when the last frame on a page is full, a new page is started.

Switching between templates is done with NextPageTemplate (deferred) or by passing a NextTemplate ID to PageBreak (immediate). AutoNextTemplate lets a template automatically switch to another template after each page — enabling first-page vs. later-page layouts without any story-side boilerplate.

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The next section demonstrates a two-column layout produced by switching to the 'two-column' PageTemplate.

## 5. Two-Column Layout

This section is rendered in a two-column layout. The story is exactly the same flat list of Flowables — the only difference is the active PageTemplate, which now provides two LayoutFrames per page instead of one.

When the left frame fills up, the engine automatically continues in the right frame. When both frames are full, a new page is started with the same two-column template (set via AutoNextTemplate).

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## 6. Summary

The Nautilus layout engine provides a clean, composable approach to high-level PDF layout in Go:

- A flat story list — no document tree to manage
- Flowable interface — extend with custom elements
- Automatic pagination — frames and pages managed for you
- Template switching — single, two-column, or any geometry
- KeepTogether — headings always accompanied by their body

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End of layout demo.